

Xgboost ML Algorithm (Classification)

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$\hat{y}$	R2
0.52	-0.48
0.58	0.42
-	-
-	-
-	-
-	-
-	-

Dataset

$y = 0.5$

$y - \hat{y}$

Salary	Credit	Approval	R1
$\leq 50K$	B	0	-0.5
$\leq 50K$	G	1	0.5
$\leq 50K$	G	1	0.5
$\leq 50K$	B	0	-0.5
$> 50K$	G	1	0.5
$> 50K$	N	1	0.5
$\leq 50K$	N	0	-0.5

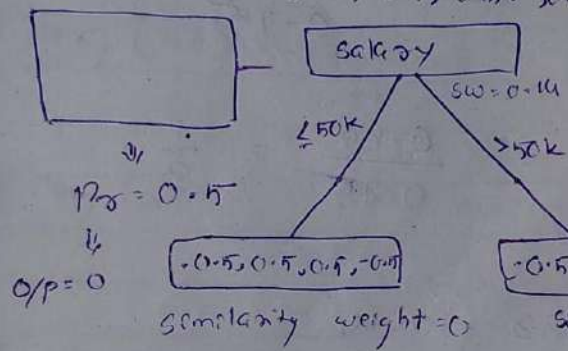
steps

- Construct a base model
- Construct a Decision Tree with root
- Calculate similarity weight

$[-0.5, 0.5, 0.5, -0.5, -0.5, 0.5, 0.5]$

$$\frac{\sum (\text{Residual})}{\sum P_2(1-P_2) + 1} = \text{cover value}$$

①



④ Calculate

similarity weight (LC) ↗ left child

$$= \frac{\sum (\text{Residual})^2}{\sum p_x (1 - p_x)}$$

$$= \frac{[(-0.5 + 0.5 + 0.5 - 0.5)^2]}{0.5(1-0.5) + 0.5(1-0.5) + 0.5(1-0.5)} = 0$$

$$= \frac{0}{1} \Rightarrow 0$$

Right child ↖

similarity weight (RC)

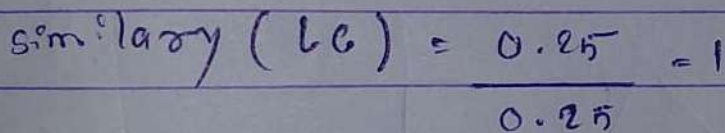
$$= \frac{(-0.5 + 0.5 + 0.5)}{0.75}$$

$$= \frac{0.25}{0.75} = \frac{1}{3}$$

Gain = $0 + 0.89 - 0.14$
 $= 0.19$

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$$7 \times 0.25 = 1.75$$



$$\text{similarity (RC)} = \frac{0.25}{0.75} = 0.33$$

$$G_{an} = 1 + 0.33 - 0$$

$$= \underline{\underline{1.33}}$$

Test data

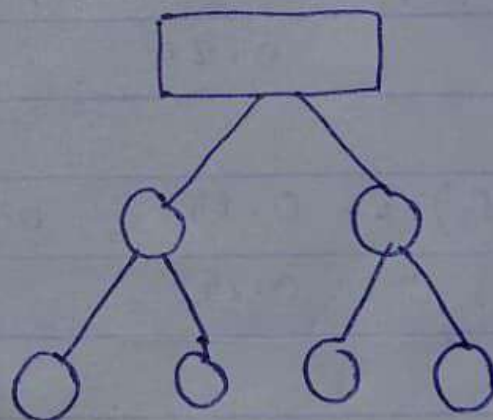
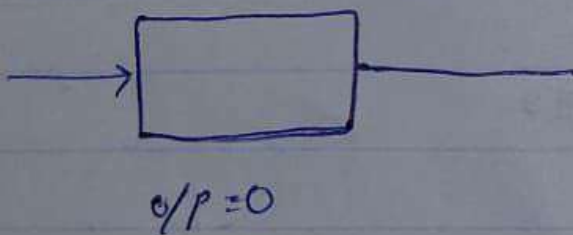
$$\log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

$$\log(\text{odds}) = \log\left(\frac{0.5}{0.5}\right)$$

$$= \log(1)$$

$$= 0$$

Final O/P



similarity weight

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$$\frac{\text{New Data}}{\text{Data}} = \sigma \left(0 + \alpha (1) \right) \quad \alpha = 0.1$$

Logistic function $\in$ Sigmoid Activation

$$= \sigma (0 + (0.1)(1))$$

$$= \frac{1}{1 + e^{-0.1}} = 0.52$$

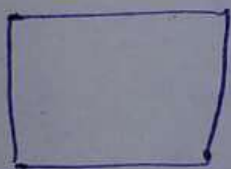
Second Record

$$o/p = \sigma (0 + \alpha (0.33))$$

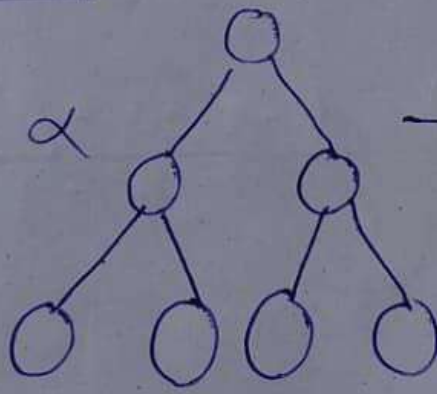
$$= \sigma (0 + 0.1 (0.33))$$

$$= \frac{1}{1 + e^{-0.033}} = \underline{0.508}$$

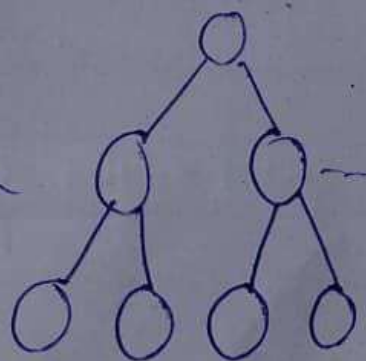
Xg boost Classifier



α



α



$$o/p = \sigma \left(\text{Base Learner} + \alpha_1 (DT1) + \alpha_2 (DT2) + \alpha_3 (DT3) \right)$$